

What Is Amazon Relational Database Service (Amazon RDS)?

Amazon Relational Database Service (Amazon RDS) is a web service that makes it easier to set up, operate, and scale a relational database in the cloud. It provides cost-efficient, resizeable capacity for an industry-standard relational database and manages common database administration tasks.

Topics

- [Amazon RDS Components](#)
- [Available RDS Interfaces](#)
- [How You Are Charged for Amazon RDS](#)
- [Monitoring an Amazon RDS DB Instance](#)
- [What's Next?](#)

Why would you want a managed relational database service? Because Amazon RDS takes over many of the difficult or tedious management tasks of a relational database.

- When you buy a server, you get CPU, memory, storage, and IOPS, all bundled together. With Amazon RDS, these are split apart so that you can scale them independently. So, for example, if you need more CPU, less IOPS, or more storage, you can easily allocate them.
- Amazon RDS manages backups, software patching, automatic failure detection, and recovery.
- In order to deliver a managed service experience, Amazon RDS does not provide shell access to DB instances, and it restricts access to certain system procedures and tables that require advanced privileges.
- You can have automated backups performed when you need them, or create your own backup snapshot. These backups can be used to restore a database, and the Amazon RDS restore process works reliably and efficiently.
- You can get high availability with a primary instance and a synchronous secondary instance that you can failover to when problems occur. You can also use MySQL, MariaDB, or PostgreSQL Read Replicas to increase read scaling.
- You can use the database products you are already familiar with: MySQL, MariaDB, PostgreSQL, Oracle, Microsoft SQL Server, and the new, MySQL-compatible Amazon Aurora DB engine (for information, see [Aurora on Amazon RDS](#)).
- In addition to the security in your database package, you can help control who can access your RDS databases by using AWS IAM to define users and permissions. You can also help protect your databases by putting them in a

virtual private cloud.

To begin learning more:

- If you are new to RDS but you are familiar with other Amazon Web Services, start with an introduction to the [Amazon RDS Components](#). This section discusses the key components of Amazon RDS and how they map to those that you currently work with on your local network.
- For an overview of all AWS products, see [What is Cloud Computing?](#)
- Amazon Web Services provides a number of database services. For guidance on which service is best for your environment, see [Running Databases on AWS](#)

Amazon RDS Components

Topics

- [DB Instances](#)
- [Regions and Availability Zones](#)
- [Security Groups](#)
- [DB Parameter Groups](#)
- [DB Option Groups](#)

DB Instances

The basic building block of Amazon RDS is the DB instance. A DB instance is an isolated database environment in the cloud. A DB instance can contain multiple user-created databases, and you can access it by using the same tools and applications that you use with a stand-alone database instance. You can create and modify a DB instance by using the Amazon AWS command line interface, the Amazon RDS API, or the AWS Management Console.

Each DB instance runs a DB engine. Amazon RDS currently supports the MySQL, MariaDB, PostgreSQL, Oracle, and Microsoft SQL Server DB engines. Each DB engine has its own supported features, and each version of a DB engine may include specific features. Additionally, each DB engine has a set of parameters in a DB parameter group that control the behavior of the databases that it manages.

The computation and memory capacity of a DB instance is determined by its DB instance class. You can select the DB instance that best meets your needs. If your needs change over time, you can change DB instances. For information about DB instance classes, see the DB Instance Class section. For pricing information on DB instance classes, go to the Pricing section of the [Amazon Relational Database Service](#)

[\(Amazon RDS\)](#) product page.

For each DB instance, you can select from 5 GB to 6 TB of associated storage

capacity. Each DB instance class has minimum and maximum storage requirements for the DB instances that are created from it. It's important to have sufficient storage so that your databases have room to grow and that features for the DB engine have room to write content or log entries.

DB instance storage comes in three types: Magnetic, General Purpose (SSD), and Provisioned IOPS (SSD). They differ in performance characteristics and price, allowing you to tailor your storage performance and cost to the needs of your database. For a complete discussion of the different volume types, see the topic [Amazon EBS Volume Types](#).

You can run a DB instance on a virtual private cloud using Amazon's Virtual Private Cloud (VPC) service. When you use a virtual private cloud, you have control over your virtual networking environment: you can select your own IP address range, create subnets, and configure routing and access control lists. The basic functionality of Amazon RDS is the same whether it is running in a VPC or not; Amazon RDS manages backups, software patching, automatic failure detection, and recovery. There is no additional cost to run your DB instance in a VPC. For more information on VPC and RDS, see [Virtual Private Clouds \(VPCs\) and Amazon RDS](#).

Regions and Availability Zones

Amazon cloud computing resources are housed in highly available data center facilities in different areas of the world (for example, North America, Europe, or Asia). Each data center location is called a region.

Each region contains multiple distinct locations called Availability Zones, or AZs. Each Availability Zone is engineered to be isolated from failures in other Availability Zones, and to provide inexpensive, low-latency network connectivity to other Availability Zones in the same region. By launching instances in separate Availability Zones, you can protect your applications from the failure of a single location. For a list of regions and Availability Zones, see [Regions and Availability Zones](#).

You can run your DB instance in several Availability Zones, an option called a Multi-AZ deployment. When you select this option, Amazon automatically provisions and maintains a synchronous standby replica of your DB instance in a different Availability Zone. The primary DB instance is synchronously replicated across Availability Zones to the standby replica to provide data redundancy, failover support, eliminate I/O freezes, and minimize latency spikes during system backups.

Security Groups

A security group controls the access to a DB instance. It does so by allowing access to IP address ranges or Amazon EC2 instances that you specify.

Amazon RDS uses DB security groups, VPC security groups, and EC2 security groups.

In simple terms, a DB security group controls access to a DB instance that is not in a VPC, a VPC security group controls access to a DB instance inside a VPC, and an Amazon EC2 security group controls access to an EC2 instance and can be used with a DB instance. For more information about security groups, see [Amazon RDS Security Groups](#).

DB Parameter Groups

You manage the configuration of a DB engine by using a DB parameter group. A DB parameter group contains engine configuration values that can be applied to one or more DB instances of the same instance type. Amazon RDS applies a default DB parameter group if you don't specify a DB parameter group when you create a DB instance. The default group contains defaults for the specific database engine and instance class of the DB instance.

DB Option Groups

Some DB engines offer tools that simplify managing your databases and making the best use of your data. Amazon RDS makes such tools available through option groups. Examples of available options are Oracle Application Express (APEX), SQL Server Transparent Data Encryption, and MySQL memcached support. For more information on option groups, see [Working with Option Groups](#).

Available RDS Interfaces

Topics

- [Amazon RDS Console](#)
- [Command Line Interface](#)
- [Programmatic Interfaces](#)

There are several ways that you can interact with Amazon RDS.

Amazon RDS Console

The Amazon RDS console is a simple web-based user interface. From the console, you can perform almost all tasks you need to do from the RDS console with no programming required. To access the Amazon RDS console, sign in to the AWS Management Console and open the Amazon RDS console at <https://console.aws.amazon.com/rds/>.

Command Line Interface

Amazon AWS provides a command line interface that gives you access to much of the functionality that is available in the Amazon RDS API. For more information, see the [AWS Command Line Interface Documentation](#) and [AWS Command Line](#)

[Reference for Amazon RDS.](#)

Programmatic Interfaces

The following table lists the resources that you can use to access Amazon RDS programmatically.

Resource	Description
AWS SDKs	The AWS SDKs include sample code, libraries, tools, documentation, and templates. To download the AWS SDKs, go to AWS Software Development Kits (SDKs) .
Libraries	AWS provides libraries, sample code, tutorials, and other resources for software developers who prefer to build applications using language-specific APIs instead of Amazon Relational Database Service's SOAP and Query APIs. These libraries provide basic functions (not included in Amazon Relational Database Service's SOAP and Query APIs), such as request authentication, request retries, and error handling so you can get started more easily. Libraries and resources are available for the following languages: • Java • PHP • Python • Ruby • Windows and .NET For libraries and sample code in all languages, see Sample Code & Libraries.
Amazon RDS API	If you prefer, you can code directly to the Amazon RDS API. For more information, see Amazon RDS API , and see the Amazon Relational Database Service API Reference .

How You Are Charged for Amazon RDS

When you use Amazon RDS, you pay only for what you use, and there are no minimum or setup fees. You are billed according to the following criteria.

- Instance class – Pricing is based on the class (e.g., micro, small, large, xlarge) of the DB instance consumed.
- Running time – You are billed by the instance-hour, which is equivalent to a single instance running for an hour. For example, both a single instance running for two hours and two instances running for one hour consume 2 instance-hours. If a DB instance runs for only part of an hour, you are charged

for a full instance-hour.

- **Storage** – The storage capacity that you have provisioned to your DB instance is billed per GB per month. If you scale your provisioned storage capacity within the month, your bill will be pro-rated.
- **I/O requests per month** – Total number of storage I/O requests that you have made in a billing cycle.
- **Backup storage** – Backup storage is the storage that is associated with automated database backups and any active database snapshots that you have taken. Increasing your backup retention period or taking additional database snapshots increases the backup storage consumed by your database. Amazon RDS provides backup storage up to 100% of your provisioned database storage at no additional charge. For example, if you have 10 GB-months of provisioned database storage, we will provide up to 10 GB-months of backup storage at no additional charge. Most databases require less raw storage for a backup than for the primary dataset, so if you don't keep multiple backups, you will never pay for backup storage. Backup storage is free only for active DB instances.
- **Data transfer** – Internet data transfer in and out of your DB instance.

In addition to regular RDS pricing, you can purchase reserved DB instances.

Reserved DB instances let you make a one-time up-front payment for a DB instance and reserve the DB instance for a one- or three-year term at significantly lower rates. For more information on reserved DB instances, see [Working with Reserved DB Instances](#)

For Amazon RDS pricing information, see the [Amazon RDS product page](#).

Monitoring an Amazon RDS DB Instance

There are several ways that you can track the performance and health of a DB instance. You can use the free Amazon CloudWatch service to monitor the performance and health of a DB instance; performance charts are shown in the Amazon RDS console. You can subscribe to Amazon RDS events to be notified when changes occur with a DB instance, DB Snapshot, DB parameter group, or DB security group. For more information about Amazon CloudWatch, see [Viewing DB Instance Metrics](#). For more information on Amazon RDS event notification, see [Using Amazon RDS Event Notification](#)

What's Next?

This section introduced you to the basic infrastructure components that RDS offers. What should you do next?

Getting Started

Create a DB instance using instructions in the [Getting Started with Amazon RDS](#) section.

Database Engine Specific Topics

You can review information specific to a particular DB engine in the following sections:

- [Oracle on Amazon RDS](#)
- [MySQL on Amazon RDS](#)
- [Microsoft SQL Server on Amazon RDS](#)
- [PostgreSQL on Amazon RDS](#)
- [Aurora on Amazon RDS](#)
- [MariaDB on Amazon RDS](#)

[Document Conventions](#)

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